PROJECT REPORT

ON

MULTI-DOCTOR OUTPATIENT CLINIC MANAGEMENT SYSTEM

(CAREPULSE OPD SUITE)

A project report submitted in partial fulfilment of the requirements
for the course / capstone project

Submitted by

Name: _____

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Under the guidance of

Guide Name: _____

Department: _____

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Academic Year: _____

Certificate

This is to certify that the project report entitled “Multi-Doctor Outpatient Clinic Management System” is a record of the project work prepared by the student named below, under the guidance of the undersigned. The report is submitted as part of the academic requirements of the course / capstone project.

Student Name: _____ **USN:** _____

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Date: _____ **Place:** _____

Declaration

I hereby declare that this report entitled “Multi-Doctor Outpatient Clinic Management System” has been prepared by me for academic purposes based on the supplied project presentation and the work carried out as part of the project. The content has been organized to describe the proposed system, its modules and architecture. Any design considerations not explicitly confirmed in the presentation are identified accordingly.

Student Name: _____

Signature: _____

Date: _____

Acknowledgement

I express my sincere gratitude to the project guide, faculty members and department for their guidance and support during the preparation of this project. I also thank my teammates and everyone who provided suggestions and feedback. Finally, I acknowledge the project presentation used as the primary source for documenting the CarePulse OPD Suite.

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+ ABSTRACT

Summary of the CarePulse OPD Suite and the scope of this report

The Multi-Doctor Outpatient Clinic Management System, presented as the CarePulse OPD Suite, is a full-stack outpatient clinic architecture designed to coordinate the activities of multiple departments in a
clinic. The solution focuses on synchronizing patient registration, live token queuing, doctor consultations, digital prescriptions, pharmacy dispensing, billing and patient history lookup.

The presentation identifies common operational difficulties in outpatient clinics: manual queue handling can create long waiting times and overcrowded reception areas; handwritten prescriptions can be difficult to interpret and may contribute to medicine or dosage errors; and disconnected cashier records can make itemized medicine billing difficult. CarePulse proposes an integrated workflow in which the departments work with shared patient and visit information.

The patient lifecycle begins at reception, where a patient record and daily token are created. The waiting-room display presents doctor-wise queue information. At the doctor desk, clinical details and vitals are recorded and a digital prescription is prepared. The prescription is then made available to the pharmacy and cashier module for dispensing, invoice calculation and receipt generation. Historical visit search and daily OPD analytics provide additional operational visibility.

The deck identifies AngularJS with ngRoute, Supabase PostgreSQL, Tailwind CSS and Row Level Security (RLS) as technical highlights. It also describes shared AngularJS services and Supabase Realtime synchronization across independent departmental views. This report documents the system as described in the presentation. Where implementation details are not specified in the deck, they are identified as design considerations rather than confirmed features.

+ SECTION 1: INTRODUCTION

Background, intent and scope of the system

Outpatient departments handle a continuous flow of patients who may visit different doctors and require registration, consultation, medicines and payment. When these activities are handled through separate paper records or disconnected tools, staff may need to repeat data entry or communicate information manually. This can slow down service and make it harder to maintain a consistent view of a patient's visit.

The Multi-Doctor Outpatient Clinic Management System is intended to bring the major OPD activities into one connected workflow. The CarePulse concept links the reception desk, live queue display, doctor desk and pharmacy/cashier operations. Each department uses the information relevant to its work while the overall patient journey remains connected.

The system is designed for a multi-doctor environment. Tokens are separated by consulting room or doctor queue, while the patient record and visit history remain accessible to authorized workstations. The presentation describes the project timeline as a three-week sprint and identifies Supabase PostgreSQL with RLS as the backend engine.

This report explains the problem, objectives, users, modules, workflow, data organization, technology choices, expected validation and possible future improvements. It follows the project content shown in the supplied capstone presentation.

+ SECTION 2: PROBLEM STATEMENT

Operational bottlenecks in outpatient clinics

Many outpatient clinics rely on fragmented processes for registration, queue management, consultation notes, prescriptions and billing. The presentation highlights three operational bottlenecks. First, manual queuing can lead to long waiting-room delays and overcrowded reception desks. Second, paper prescriptions may be difficult to read and can contribute to medicine or dosage errors. Third, siloed cashier records can prevent staff from accessing accurate, itemized medicine billing information.

These disconnected processes make it difficult to maintain a continuous patient journey across departments. A patient may be registered at reception, wait for a doctor, receive a prescription and then visit the pharmacy or cashier. If each stage uses separate records, staff must rely on manual communication and repeated entry.

Problem statement: To design an integrated multi-doctor outpatient clinic management system that coordinates patient registration, doctor-wise token queues, clinical consultation, digital prescriptions, pharmacy dispensing, billing and historical visit lookup through a shared data platform.

+ SECTION 3: AIM AND OBJECTIVES

Project aim, derived objectives and the stated target

Aim: To develop an integrated outpatient clinic management system that improves coordination among reception, doctors, pharmacy and cashier operations in a multi-doctor clinic.

The objectives derived from the presentation are:

1. Create patient records with unique Medical Record Number (MRN) identifiers;
2. Generate sequential daily tokens for doctor queues;
3. Display queue status and support one-click token progression with audio alerts;
4. Record patient vitals, complaints and diagnoses during consultation;
5. Create structured digital prescriptions with medicine rows and dosage instructions;
6. Provide allergy warning safeguards;
7. Route completed prescriptions to pharmacy operations;
8. Calculate itemized invoices with applicable tax and support cash, UPI and card payment modes;
9. Provide printable header slips or receipts;
10. Enable historical visit lookup using MRN or phone number; and
11. Present daily OPD performance information.

The deck states a goal of reducing patient wait times by 60%. This is a project target presented in the slides, not a measured result established by the supplied material.

+ SECTION 4: EXISTING APPROACH AND PROPOSED SYSTEM

Fragmented workflow versus the integrated CarePulse approach

4.1 Existing Approach

The presentation characterizes the existing workflow as fragmented. Queue handling is manual, prescriptions are paper-based, and cashier records are siloed. These practices can create delays, legibility problems and incomplete access to itemized billing information.

4.2 Proposed Approach

CarePulse integrates the core departments through a shared application and database. Reception creates the patient record and daily token. The queue display shows doctor-wise token status. The doctor records clinical information and prepares a digital prescription. Pharmacy and cashier staff use the prescription line items to dispense medicines and calculate the invoice. Historical visits can be searched across the departmental workstations.

The proposed system is not simply a replacement for paper forms. Its central idea is interdepartmental synchronization: a change made during one stage should become available to the next relevant stage without requiring the same information to be entered again. The presentation identifies shared AngularJS services and Supabase Realtime as the synchronization approach.

+ SECTION 5: USERS AND DEPARTMENTAL RESPONSIBILITIES

Roles and responsibilities across clinic departments

User / Department	Responsibilities
Reception staff	Register patients, generate or retrieve MRNs, identify possible duplicate phone numbers and issue daily tokens. The deck describes an automated MRN generator, duplicate detection and sequential daily token format such as T-014.
Waiting-room staff and patients	View doctor-wise token queues and understand whether a token is waiting, in consultation or complete. The display is intended to update in real time and provide audio chime alerts when a token is called.
Doctors and clinical staff	Review patient history, record blood pressure, pulse, temperature and weight, enter chief complaints and diagnoses, and prepare digital prescriptions. They can add or remove medicine rows, select dosage timing flags and receive allergy warning popups.
Pharmacy staff	View prescriptions routed from completed consultations, review medicine line items and support dispensing. The pharmacy workflow is linked to invoice calculation.
Cashier staff	Access consultation fees, medicine totals and tax information, calculate the amount payable and issue a printable receipt. The slide lists cash, UPI and card as payment modes.
Clinic management	Review daily OPD indicators such as total daily revenue, queue volume and the automatic mock fallback/zero-fail metric shown in the presentation.

+ SECTION 6: SYSTEM MODULES

Module-by-module description of the CarePulse OPD Suite

6.1 Patient Registration and MRN Management

The reception registration module creates a patient record and assigns a unique MRN. It also checks for phone-number collisions during check-in to flag possible duplicate registrations.

6.2 Daily Token Generation

The system issues sequential daily tokens, illustrated in the deck with a T-001 to T-999 format. Tokens are associated with the relevant doctor queue so that multiple consulting rooms can operate independently.

6.3 Live Queue Display

Separate token screens are provided for consulting rooms. A Call Next control advances the queue, and audio chimes notify users. Status markers include Waiting, In Consultation and Complete. The presentation lists an average waiting-time target of less than 12 minutes and real-time synchronization latency below 200 ms as design indicators.

6.4 Clinical Examination

The doctor desk captures vitals including BP, pulse, temperature and weight, along with chief complaints and clinical diagnoses. The interface also provides access to historical visit records.

6.5 Dynamic Prescription Builder

Doctors can add or remove medicine rows, select medicine details and configure dosage timing flags such as 1-0-1, 1-1-1 and SOS. Allergy warning popups are identified as a safety safeguard. Completed consultations automatically route prescriptions to the Pharmacy Module.

6.6 Pharmacy and Cashier

The invoice includes consultation fees, pharmacy line items and clinic/statutory tax. The sample slide illustrates a consultation fee, Amoxicillin 500 mg capsules and a 5% tax line. It also shows a net transaction total of ₹651.00 for the example displayed. Cash, UPI and card payment modes are listed.

6.7 Historical Visit Search and EHR Lookup

Staff can search using a patient’s phone number or MRN. The slide illustrates historical visits with dates, doctor names, diagnoses and medicines. It emphasizes access to longitudinal patient history across the four departmental workstations.

6.8 Reports and Performance

The analytics slide shows sample indicators of total daily revenue ₹42,850 and 68 tokens. It also displays an automatic mock fallback/zero-fail indicator. These are presentation examples and should not be treated as verified live operational results.

+ SECTION 7: WORKFLOW AND PATIENT LIFECYCLE

Four-stage patient journey across departments

The project presentation defines a four-stage patient lifecycle: Reception Desk → Live Queue Display → Doctor Desk → Pharmacy & Billing.

→ **Reception Desk** → **Live Queue Display** → **Doctor Desk** **Pharmacy & Billing**

Stage	Description
Stage 1 — Reception	The patient is registered or matched to an existing record. A unique MRN is used as the patient identifier. The system checks for duplicate phone numbers and issues a sequential daily token for the appropriate doctor queue.
Stage 2 — Live queue	The token appears on the consulting room's queue display. When the doctor is ready, staff use Call Next to advance the queue. The display updates the token status and can play an audio chime. Status values shown in the deck are Waiting, In Consultation and Complete.
Stage 3 — Doctor consultation	The doctor opens the patient record, reviews available history, records vitals and clinical details, and prepares a digital prescription. Medicine rows and dosage instructions are entered in a structured form. Allergy warnings are intended to provide an additional safety check.
Stage 4 — Pharmacy and billing	The completed prescription is routed to the pharmacy module. Medicine line items, consultation fees and tax are used to calculate the invoice. The cashier can record a supported payment mode and produce a printable receipt.

After the visit, the record contributes to the patient's historical visit history and clinic reporting. The workflow is intended to reduce duplicate entry and keep the departments synchronized.

+ SECTION 8: SYSTEM ARCHITECTURE

Technology stack and component interaction

The deck presents a full-stack outpatient clinic architecture with AngularJS as the front-end application framework and Supabase PostgreSQL as the backend database. AngularJS ngRoute is listed among the technical highlights. Tailwind CSS is identified for interface styling. Supabase Realtime is used in the stated synchronization concept, while Row Level Security (RLS) is listed as a security-related technical highlight.

The application is divided into independent departmental views: reception, queue display, doctor desk, and pharmacy/cashier. Shared AngularJS services are described as the client-side mechanism for sharing application state. Supabase Realtime is described as the mechanism supporting synchronization across the views.

At a conceptual level, the front end sends and retrieves information through the backend data layer. Patient demographics, visit and queue information, prescriptions and invoices are represented as related entities. The same patient identity links the different stages of the visit.

The supplied deck does not provide a complete deployment diagram, API specification, authentication flow or source-code listing. Those implementation details should be added from the actual project code if the report is to be used as final technical documentation.

+ SECTION 9: DATABASE DESIGN AND ENTITY ORGANIZATION

Core entities and their relationships

The relational schema slide identifies four main entities: patients, visits, prescriptions and invoices. These entities represent the core information required to coordinate the outpatient workflow.

Entity	Description
Patients	Stores primary demographic information and uses a unique MRN as the patient identifier. The MRN supports patient lookup and links the person to their visits.
Visits	Tracks token numbers, live doctor queues, symptoms and vitals. A visit represents a patient's encounter on a particular date and supports queue and clinical workflow information.
Prescriptions	Stores dynamic line-item medicine information. The slide indicates that medicine repeaters are stored inside JSONB columns. The sample data visual includes a patient identifier, doctor and a medicines array.
Invoices	Stores consultation fees, medicine totals, tax and payment modes. The invoice is associated with the visit and supports itemized billing.

The presentation does not specify all column names, primary/foreign key definitions, indexes, constraints or complete JSONB structure. A final implementation report should include the actual Supabase table definitions and relationships. A possible logical relationship is one patient to many visits, one visit to a prescription, and one visit to an invoice; this relationship should be verified against the implemented database.

+ SECTION 10: USER INTERFACE AND SCREEN DESIGN

Department-specific interfaces

The user interface is organized around the tasks performed by each department. Reception has a registration interface for entering patient information and checking duplicates. The queue screen separates tokens by consulting room and provides a Call Next action with visible status markers.

The doctor desk combines clinical examination details with a prescription builder. The presentation shows a clinical examination suite and a prescription panel containing medicine rows. The design supports adding or removing medicine entries and displaying dosage instructions.

The pharmacy and cashier interface presents invoice line items, quantities or days, unit prices and subtotals. It provides a net transaction total and printable header slips. The historical record interface supports multi-parameter lookup and displays past visits with date, doctor, diagnosis and medicines.

The analytics interface uses summary cards to display daily revenue, queue volume and an automatic mock fallback indicator. Tailwind CSS is named in the deck as the styling technology. The actual screen layouts and visual details should be documented with screenshots from the running application.

+ SECTION 11: NON-FUNCTIONAL REQUIREMENTS AND SAFETY CONSIDERATIONS

Performance, usability, data consistency, access control and clinical safety

Requirement	Description
Performance	The presentation gives real-time synchronization latency below 200 ms as a target and average waiting time below 12 minutes as an operational indicator. These should be tested under realistic network and workload conditions.
Usability	Department-specific screens should make common tasks clear and reduce unnecessary navigation. Reception, doctors and pharmacy/cashier staff should see the information relevant to their responsibilities.
Data consistency	A patient identifier should connect registration, visits, prescriptions and invoices. Updates to queue status and completed prescriptions should be reflected in the relevant views.
Access control	Row Level Security (RLS) is listed among the technical highlights. The actual access policies should be checked to ensure users can access only the records and actions required for their role.
Clinical safety	Allergy warnings and structured dosage timing flags are included in the presentation. These are software safeguards and should not replace professional clinical judgment. Medicine selection, dosage and allergy data must be verified by qualified clinical staff.
Privacy and confidentiality	Patient demographics and health information are sensitive. The deck does not specify a full privacy policy, backup strategy, encryption configuration or compliance assessment. These should be confirmed and documented before real patient data is used.

+ SECTION 12: TESTING AND VALIDATION PLAN

Proposed validation checks — not completed test results

The presentation does not include a detailed test report or test-case results. The following checks are therefore a proposed validation plan, not claims that testing has already been completed.

Test Area	Checks
Registration tests	Verify that a new patient receives a unique MRN, required fields are validated, and a duplicate phone number is flagged for review.
Token tests	Verify sequential daily token generation, correct doctor queue assignment, and correct status progression from Waiting to In Consultation to Complete.
Realtime tests	Open departmental views simultaneously and verify that token calls and status changes appear across the relevant screens. Measure synchronization delay under normal and peak loads.
Clinical and prescription tests	Verify that vitals and diagnoses can be recorded, medicine rows can be added or removed, dosage flags are stored correctly, and allergy warnings appear when configured allergy information matches a selected medicine.
Billing tests	Check consultation fee, medicine quantity, unit price, subtotal and tax calculations. Verify the displayed total and confirm that cash, UPI and card payment modes can be selected and recorded.
History and analytics tests	Search by MRN and phone number, confirm that the correct historical visits are displayed, and reconcile daily revenue and token totals with the underlying records.
Security tests	Review Supabase RLS policies using users with different roles. Confirm that unauthorized access to patient, prescription and invoice records is blocked.

+ SECTION 13: LIMITATIONS AND FUTURE ENHANCEMENTS

Documentation gaps and possible extensions

The supplied presentation is a capstone defense deck and does not document every implementation detail. It does not provide full source code, complete database definitions, deployment architecture, detailed user-role permissions, backup and recovery procedures, or formal test results. These details should be added from the implemented system.

Future enhancements may include appointment scheduling, SMS or WhatsApp reminders, patient self-service access, doctor availability management, medicine inventory tracking, low-stock alerts, advanced analytics, downloadable reports and automated backup monitoring. These are suggested extensions and are not represented as existing features in the supplied deck.

Additional improvements could include audit logs for sensitive record changes, configurable role permissions, stronger validation of clinical data, and a formal review of privacy and security requirements before deployment in a real clinic.

+ SECTION 14: CONCLUSION

Summary of contributions and caveats

The Multi-Doctor Outpatient Clinic Management System, CarePulse OPD Suite, proposes an integrated way to coordinate the main activities of a multi-doctor outpatient clinic. It connects reception registration, daily token queuing, doctor consultation, digital prescriptions, pharmacy dispensing, cashier billing and historical patient record lookup.

The key contribution described in the presentation is interdepartmental synchronization through shared AngularJS services and Supabase Realtime. The relational data model is organized around patients, visits, prescriptions and invoices. The deck also identifies AngularJS ngRoute, Supabase PostgreSQL, Tailwind CSS and Row Level Security as core technical elements.

By reducing fragmented records and repeated manual communication, the system aims to improve the flow of information between departments and support a more organized patient experience. The stated 60% waiting-time reduction and the performance figures shown in the slides should be treated as project goals or sample indicators until validated through actual deployment and testing.

+ REFERENCES AND APPENDIX

Source material, terminology and sample values

Primary source: “Multi-Doctor Outpatient Clinic Management System — Capstone 2 Deck” (10-slide PDF presentation supplied for this report).

Technology names documented in the presentation: AngularJS / ngRoute, Supabase PostgreSQL, Supabase Realtime, Tailwind CSS and Row Level Security (RLS). This report does not add external technical claims beyond the project deck.

Appendix A — Terms

Term	Meaning
OPD	Outpatient Department
MRN	Medical Record Number
EHR	Electronic Health Record
RLS	Row Level Security
JSONB	PostgreSQL binary JSON data type
UPI	Unified Payments Interface

Appendix B — Sample Token Format

T-001 to T-999. Sample queue states: Waiting, In Consultation and Complete. Sample dosage timing flags: 1-0-1, 1-1-1 and SOS.

Appendix C — Sample Indicators Shown in the Deck

Daily revenue ₹42,850; queue volume 68 tokens; average waiting time target below 12 minutes; real-time synchronization target below 200 ms. These are reproduced as slide examples/targets, not independently verified results.